MIL-C-24643/43B
3 October 1986
SUPERSEDING
MIL-C-0024643/43A(SH)
8 November 1985
MIL-C-24643/43
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPES LS2SJ, LS3SJ, AND LS4SJ (INCLUDING VARIATIONS LS2SJA, LS3SJA, AND LS4SJA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSSJ (unarmored) and LSSJA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, in accordance with ASTM B 286, sizes 14 and smaller. Uncoated for AWG sizes 12, class B stranding, and larger.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation (see table I for dimensions). Standard identification code applied by method 3.
Fourth - Two, three or four conductors, as required, shall be cabled together with fillers as necessary to form a firm, round assembly. (See table I for maximum length of lay).
Fifth - Braided shield of tin coated copper strands AWG no. 30, or smaller, providing a minimum coverage of 85 percent.
Sixth - Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LSSJ

Seventh - Cross-linked polyolefin jacket with cable surface marking.

Construction variant for type LSSJA

Seventh - Cross-linked polyolefin jacket.
Eighth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/43	Conductor				Diameter over insulation (nom) (inch)	Lay of conductor (max) (inches)	Overall diameter (inches)				Conductor resistance per 1000 feet at 25°C max (ohms)
			Size 3/				Type LSSJ		Type LSSJA		
	Size 2/	No.	ASTM B 286	ASTM B 8			min	max	min	max	
-01UO	22	2	22-19		0.067	1-1/2	0.261	0.275	0.311	0.325	16.54
-02UO	20	2	20-19		.073	1-1/2	.273	.290	.323	.340	10.16
-03UO	18	2	18-19		.084	1-1/2	.295	.310	.345	.360	6.47
-04UO	16	2	16-19		.091	2-1/2	.309	.325	.359	.375	5.02
-05UO	14	2	14-19		.105	2-1/2	.337	.350	.387	.400	3.17
-06UO	12	2		12 (class B)	.145	3	.417	.430	.467	.480	1.72
-07UO	11	2		10 (class B)	.160	3	.447	.460	.495	.510	1.085
-08UO	9	2		9 (class B)	.200	4	.525	.545	.575	.595	.86
-09UO	7	2		7 (class B)	.235	4	.600	.615	.650	.665	.54
-10UO	22	3	22-19		.067	1-1/2	.271	.285	.321	.335	16.54
-11UO	20	3	20-19		.073	1-1/2	.284	.300	.334	.350	10.16
-12UO	18	3	18-19		.084	1-1/2	.308	.325	.358	.375	6.47
-13UO	16	3	16-19		.091	2-1/2	.323	.340	.373	.390	5.02
-14UO	14	3	14-19		.105	2-1/2	.353	.370	.403	.420	3.17
-15UO	12	3		12 (class B)	.145	3	.440	.455	.490	.505	1.72
-16UO	9	3		9 (class B)	.200	4	.594	.620	.644	.670	.86
-17UO	20	4	20-19		.073	1-1/2	.303	.320	.353	.370	10.16
-18UO	16	4	16-19		.091	2-1/2	.346	.360	.396	.410	5.02
-19UO	14	4	14-19		.105	3	.380	.395	.430	.445	3.17

1/ Unarmored, overall shield. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSSJ and LSSJA.
3/ Size designation does not correlate to conductor size for all types and sizes.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	(see table I)
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	1200
Conductor to shield	600
Shield to water	600
Insulation resistance - megohms/1000 feet, minimum	500
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50

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EXAMINATION AND TESTS: (Continued)

	Requirements
Shrinkage	No failure
Permanence of printing (jacket) type LSSJ only - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Armor (type LSSJA only) - conformance to material construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Aging and compatibility (cable) (125 ± 5°C)	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum	75
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Flame propagation (cable)	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
	Smoke index, maximum	
	Jacket	25
	Fillers	—
(B)	Insulation	45
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 1000 feet (nominal)

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-43)

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